

# Project Development Phase

## Data Processing Cycle & Core Business Questions

|               |                                      |
|---------------|--------------------------------------|
| Date          | 02 July 2026                         |
| Team ID       | SWTID-2026-4978                      |
| Project Name  | Strategic Product Placement Analysis |
| Maximum Marks | 4 Marks                              |

### 1. Data Processing Steps

The lifecycle of converting the raw transactional logs into verified, interactive dashboards follows a defined execution sequence:

- **Step 1: Data Connection & Schema Validation:** Connect the local environment directly to the unorganized transaction repository (Product Positioning.csv). Validate data data types, confirming coordinate points match floor parameters perfectly.
- **Step 2: Attribute Segregation:** Map out independent structural variables including physical store sections (Aisle, End-cap, Front), product divisions (Clothing, Electronics, Food), and consumer groupings.
- **Step 3: Analytical Calculation Processing:** Implement native calculated filters to mathematically isolate campaign marketing promotional volume drops from native floor-space geometry baseline trends.
- **Step 4: Interactive Matrix Compilation:** Link individual worksheets through global cross-sheet dashboard actions so that selecting a profile updates the spatial map metrics simultaneously.

### 2. Strategic Business Questions & Visualization Mapping

The developmental dashboard answers defined operational challenges using specific target visual chart types:

| No.  | Target Business Question Addressed                                                                                            | Selected Chart / Visual                   | Target Operational Action                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-----------------------------------------------------------------------------|
| BQ-1 | Which physical real estate zones (Aisle, End-cap, Front) generate the highest overall sales velocity per category?            | Spatial Floor Treemap Diagram             | Relocate low-performing stock into verified, high-volume sectors.           |
| BQ-2 | How do regional internal prices index against local competitive market benchmarks?                                            | Side-by-Side Benchmark Bar Plots          | Adjust active category pricing gaps to protect margin limits.               |
| BQ-3 | Which specific consumer profiles (Families, Students, Seniors) hold the highest demand lift for specific inventory divisions? | Demographic Segment Donut Plots           | Tailor promotional displays directly to local neighborhood profile traffic. |
| BQ-4 | Does a sales surge genuinely stem from a marketing campaign or from permanent shelf geometric placement?                      | Promotion Lift Matrix (Text Table Matrix) | Isolate promotional costs from baseline placement value.                    |